Degree-two local collisions: exact finite checks

Result. Lean proves the normalized weighted admissible-label coefficient1-I2-I2²+I4, and checks its actual fresh two-step collision probability on the complete Z7 random-hash population with steps1,3. A second distinct/nonzero table1,2 violates the source degree condition and directly disproves removing it. This checks a known Bernstein–Lange2013 correction; no novelty or limiting collision constant is claimed. [pcc:unverified]

Weighted labels	I2 = sum p_i^2; I4 = sum p_i^4; actual admissible label sum Degree-two coefficient = 1 - I2 - I2² + I4
Actual good table	Z7; steps1,3; 128static hashes x49ordered start pairs One-step count448/rate1/14; fresh two-step336/rate3/56
Hypothesis control	Steps1,2: still distinct/nonzero, violates degree condition Fresh two-step count112/rate1/56; omitted guard false

Generic label identity. Sum the products p_i p_j p_k p_l over different final labels j,l, excluding oppositely ordered copies of the same two steps. For every finite real weight list summing to1, Lean derives1-I2-I2²+I4 by explicit label exclusions and off-diagonal finite sums. This is an actual weighted sum identity, not a coefficient assumption or native oracle. The conversion to actual graph probabilities is checked separately at the stated finite tables. [pcc:unverified]

Actual graph and probability law. The state map is x→x+g(h(x)) on Z7, with every one of the128 maps h:Z7→Fin2 and all49 ordered starts. A fresh two-step event requires all four queried states x,y,F(x),F(y) distinct and F²(x)=F²(y). Lean checks degree0/1/2 sum distinctness modulo commutativity for g=(1,3). Kernel enumeration gives one-step448 and fresh two-step336 among6272 samples. A genuine PMF has pointmass equal to the product of seven uniform vertex-label factors and two uniform-start factors. Its exact event rates are1/14 and3/56, and3/56 equals the proved uniform coefficient3/8 divided by7. [pcc:unverified]

Direct negative control. The table g=(1,2) remains injective and nonzero but fails the degree condition:1+1=2 mixes degrees. Its actual fresh two-step count is112 and rate1/56. Lean directly negates the formula after replacing the source degree qualification by injectivity/nonzero alone. The two actual good-table event rates also differ; a degree-one local coefficient does not equal the checked degree-two coefficient. No qualified source theorem is refuted. [pcc:unverified]

Boundary. The generic label identity and actual finite graph checks do not prove the general actual-group/support probability coefficient, a limiting random-increment-table firstcollision penalty or practical keyedhash behavior. The separate exact static-hash/iid firstcollision survival bridge remains unchanged. Source higher-degree asymptotic heuristic corrections are credited, not promoted to proved limiting results. [pcc:unverified]

Checks and retained failure. Whole final gate26/0 failures/0 trusted native; standard propext/Classical.choice/Quot.sound only. Complete1347-character binding4c1acf0fd9722f21066a7ce0cb3de28321e7e626a7fbb2eaa2efafd2c441c000, independent native1complete/0prefix/0unresolved. Initial native parser failed on scoped ENNReal notation; equivalent explicit type name rebuilt/regated, frozen parser unchanged. Seven build snapshots/four failed builds, both native inputs/protocols and both gates retained: 96files/44objects/12128-byte lossless XZ plus final gate files. [pcc:unverified]

Sources and assurance. Bernstein–Lange Two grumpy giants and a baby2013§§3–4, full selected printed94–97/PDF9–12 read/visually inspected; degree≤2 and distinct-query qualifications retained. Galbraithch14v2 selected288/296–297 and prior actual variance/firstcollision studies credited. Official Mathlib finite sums/PMF/ENNReal APIs checked. KB/Conductor unavailable, primary/prior/official fallback; no fullpaper-read claim. Lean4.35.0/Mathlib56a00585…, defaults; all finite counts by kernel decisions, no native evaluator/holes/custom axioms/flags/limits/waivers/new crypto experiment/spend/upload. Signed standing unverified. [pcc:unverified]